Robe variants tested, October 2026

UO_Model3D, branch model_fixing5: cloth physics, fit, fabrics and shading of one free robe model (24,366 vertices after conform)

Summary

- Goal: a loose robe that moves like cloth (the legs push it out, no fixed leg-hull target) and still sits on the original body 400 like the original UO robes. About 45 physics configurations, 3 fabrics (silk, cotton, wool), 3 fit variants and 22 shading variants were tried.
- Physics: a free Blender cloth on this fine mesh needs a weak pin to its kinematic target (goal 0.2, pin target = skinned + 0.5 x leg push), a contact margin of 0.02 m (thin margins let the legs through), vertex mass scaled to the mesh (mscale 0.033) and in-plane stiffness 5 for every fabric (stiffer fabrics are pulled up by the pin: the hem climbed 0.26 m in cotton and 0.47 m in wool in the walk; at 5 it is 0.04-0.05 m).
- Fabrics: silk, cotton and wool differ only in mass, bending and damping after that fix; they move almost alike because the pin is the anchor. Denim and leather were never run.
- Shape: the loose and the tight fit differ by 0.7-0.9 cm (0.3 px); the sleeves are about 10% thicker than the original robe 469, the skirt is about 1.7 x wider at the hem, the waist is 2 cm per side wider.
- Look: the original robe is painted with dark creases (brightness spread in the sleeve zone 54, our smooth shell 31). A contrast curve reproduces it (k 1.8: 52); light direction, own shadow, normal map strength, occlusion and a 1 px inner line did not (or looked like a drawn line).
- Decision (2026-10-20): base look of a robe / skirt = uo_contrast 1.5, uo_shade_sleeve 0.4, uo_shade_width 4 (uo_make_item.py scene_of; the recipe's own scene wins). Four full 35-action .vd files were made: silk, cotton, wool, tight silk.

1. Metrics used

Metric	Definition
hem lift	mean vertical offset (m) of the simulated hem (rest z < 0.3 m) from its kinematic target, averaged over the frames of an action; + = the hem climbed. Silk (the soft reference) gives +0.04.
leg px (item_skin_check.py)	pixels of thigh and shin (feet counted apart) in the 3D preview of body + item, over 100 frames of walk and run; split into 'through the cloth' (the leg is in front of an item pixel) and 'in holes' (no item pixel there).
holes (item_body_holes.py)	pixels of the ORIGINAL body 400 inside the closed outline of the clothing layer, i.e. what the client shows; legs only, hands and head apart.
stand-off	distance (cm) of item vertices from the skin in the rest pose, torso 1.05-1.50 m, sleeves, skirt below 0.9 m (BVH of the body).
torso band width	extra half-width (cm) of the item silhouette over the original body silhouette per row, z 1.02-1.55 m, against the original robe 469 (sprite) in the same frames.
arm zone ratio	item pixels in the arm (sleeve) or hand zone divided by the pixels of that body part; zones from body_part_raster.py labels, torso labels excluded.
brightness spread	standard deviation of luminance (0-255) of the item pixels in the sleeve or hand zone; also the 5th percentile and the share of pixels darker than (item median - 35). Original robe 469: sleeve 54, hand 52 (walk).

2. Cloth physics

Settings of the final simulation (t1): fabric=<name> mscale=0.0333 goal=0.2 pushw=0.5 gravity=1.5 in_max=-1 max_move=0 smooth=0.04 arm_goal=0.95 preroll=20 cycles=2 thick=0.02 dist=0.005 (riding: mounted=1 horse=1)

Free simulation (goal 0)

ID	Change	Result	Verdict
P1	goal 0, mass 0.3 kg per vertex, contact margin 0.02 m, clamps off	walk: largest cloth move 18.5 m, median 0.6 m	rejected: 0.3 kg x 24k vertices is 7 t of cloth; the skirt falls apart
P2	mass from fabric density (cotton 3.7e-5, silk 1.5e-5 kg per vertex), margin 0.02	largest move 1.2 / 1.27 m; the hem band (z < 0.4 m) is lifted by +0.68 m on average	rejected

Stand pose controls (hem z mean, start 0.249 m, rest 0.240 m)

ID	Change	Result	Verdict
P3	pin goal 0.4 (the old hybrid)	0.249 (stable)	reference
P4	collision off	0.250 (stable)	the collision margin is the cause of P6
P5	mass 0.3, goal 0	0.162, hem radius 0.32 -> 0.22	collapses downwards

ID	Change	Result	Verdict
P6	physical mass, margin 0.02	0.314 (hem +6.5 cm, still rising)	the collision margin lifts the free hem; margins 0.002-0.008 do not
P7	margin 0.002, dist 0.001	0.249 (stable)	legs tunnel in motion (see leg px)
P8	margin 0.008, dist 0.002	0.251	
P9	margin 0.02, quality 3	0.278	
P10	mass 0.01	0.231 (droops 2 cm)	OK
P11	physical mass, damping x 0.01	0.257	

Walk, free cloth, margin 0.006 (hem lift, m)

ID	Change	Result	Verdict
P12	cotton, physical mass	+0.28	hem climbs to the knees; accordion folds
P13	mass 0.01, damping 5 / 1	+0.24	
P14	mass 0.003, damping 1 / 0.1	+0.29	
P15	mass 0.01, damping 1 / 0.2	+0.25	
P16	bending 0.02, mass 0.01	+0.24	bending is not the cause
P17	mass 0.02, bending 0.005	+0.21	
P18	mass 0.05, tension 100	+0.26	
P19	quality 16, margin 0.006	+0.31 (10 min)	finer steps do not help
P20	quality 24, margin 0.01	+0.43 (17 min)	
P21	quality 16, margin 0.015	+0.48 (18 min)	
P22	pinned band 0.05 m (ramp)	+0.24	
P23	ramp 0.05, margin 0.015	+0.34	
P24	ramp 0.15	+0.22	
P25	pin goal 0.05	+0.19	
P26	pin goal 0.10	+0.10	
P27	pin goal 0.20	+0.02	chosen: the cloth hangs, the fabric parameters do the rest

Analysis of the free cloth

ID	Change	Result	Verdict
P28	edge lengths of the free result vs rest	median ratio 1.00 (not stretched); lowest 2% of vertices z: rest 0.16, target 0.06, sim 0.34 m	the cloth folds up like an accordion: metastable, nothing unfolds it

Pin target and weight (walk and spell, cotton)

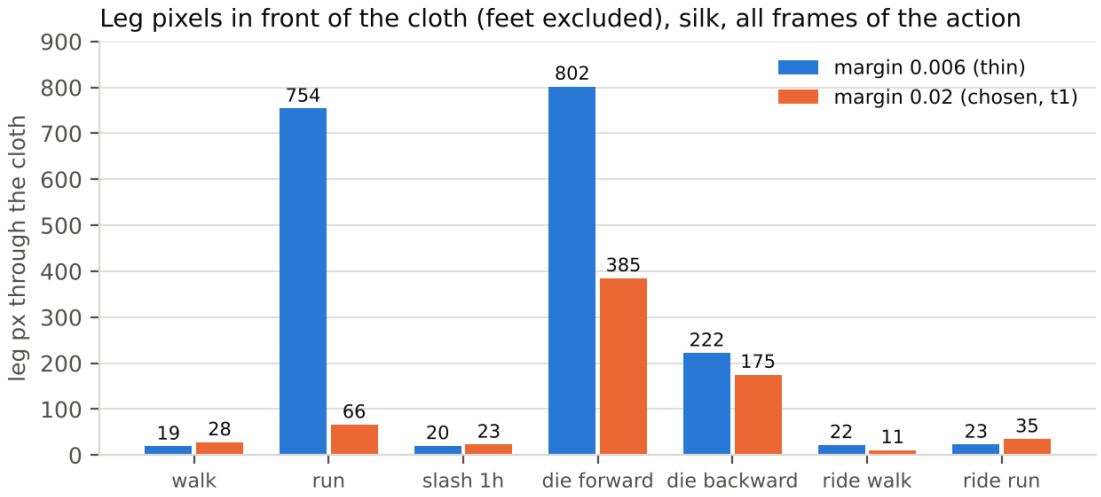
ID	Change	Result	Verdict
P29	pin target = skinned + 0.5 x leg push (pushw 0.5)	hangs straighter than the hull (pushw 1); smaller knee bulge in the spell	chosen
P30	pushw 0 (hangs from the pelvis only)	legs poke through the front of the robe	rejected
P31	pushw 0, gravity x 2	same, more exposure	rejected; gravity 1.5 kept with pushw 0.5

2.1 Leg exposure (silk, walk + run, 100 frames)

item_skin_check.py, feet excluded from the leg numbers. Total = through the cloth + in holes.

Variant	Total	Through	In holes	Note
hull target (pushw 1), margin 0.006	671	352	319	first batch of the work (hull target)
pushw 0.5, margin 0.006	2396	1360	1036	legs come out of the hem in the stride
t1: margin 0.02, dist 0.005	1435	304	1131	chosen; 7-10 x slower (22 vs 3 min per action)
t2: margin 0.015, quality 12, cq 6	2118	850	1268	
t3: margin 0.025, quality 12, cq 6	2376	761	1615	about 1 h per action
n1: single-sided collider (normals), margin 0.006	2628	1564	1064	a colleague's idea; no help
n2: single-sided + normal impulses, margin 0.006	2642	1522	1120	no help
E1: envelope after the sim, margin 0.02 (on margin 0.006)	693	196	497	flattens the hem
E2: envelope, margin 0.04, tent	492	74	418	best numbers, bell-shaped skirt again; rejected by the user as unnatural

Client view (item_body_holes.py), leg pixels of the original body inside the outline, 100 frames: pushw 0.5, margin 0.006 71; t1 (margin 0.02) 85; E2 (envelope) 35. The 3D preview exaggerates: its larger numbers include legs outside the silhouette.



Per action, silk, feet excluded (total / through the cloth): walk 49/28 (thin 31/19); run 156/66 (thin 840/754); slash 1h 23/23 (thin 20/20); die forward 409/385 (thin 976/802); die backward 212/175 (thin 270/222); ride walk 72/11 (thin 65/22); ride run 105/35 (thin 100/23). Sum: t1 1026 (723 through), thin 2302 (1862).

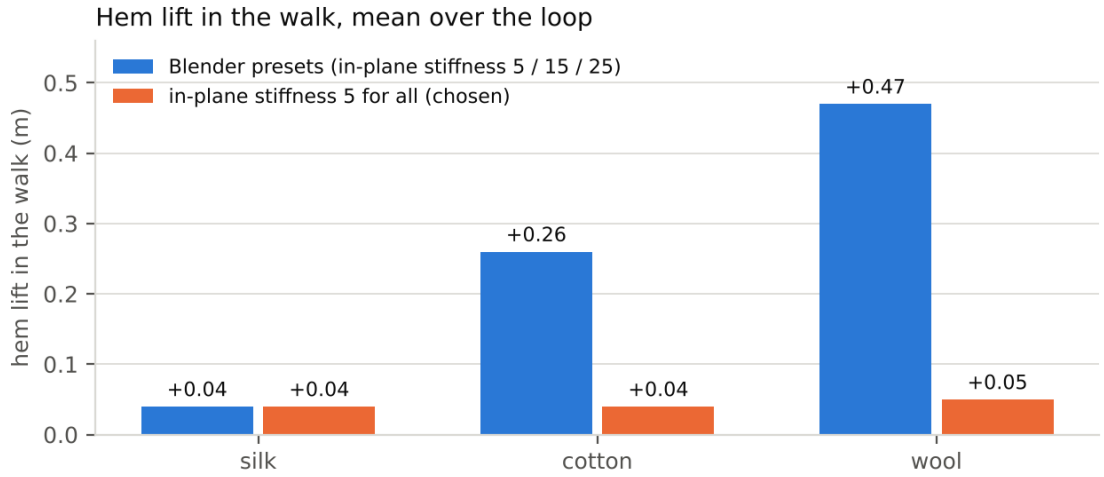
2.2 Fabrics

Blender's own presets (Cotton, Denim, Leather, Silk; wool is ours) with the vertex mass scaled by 0.0333 (mscale) for this 24k vertex mesh and the in-plane stiffness set to 5. A heavier fabric hangs lower and swings less, but the pin to the target (goal 0.2) is the anchor, so the fabrics move almost alike.

Fabric	Mass (preset)	In-plane stiffness	Bending	Damping	Status
silk	0.15	5	0.05	1	complete (35 actions)
cotton	0.3	5 (preset 15)	0.5	5	complete (35 actions)
wool	0.5	5 (ours 25)	2	10	complete (35 actions)
denim	1.0	5 (preset 40)	10	25	not run
leather	0.4	5 (preset 80)	150	25	not run

ID	Fabric test (walk)	Hem lift	Verdict
F1	cotton, tension / compression / shear 15, bending 0.5, damping 5	walk hem lift +0.26	baseline of the presets
F2	wool, 25 / 25 / 20, bending 2, damping 10	+0.47 (0.39 -> 0.53 inside the loop)	baseline of the presets
F3	wool, damping x 0.1 / x 0.033	+0.38 / +0.46	damping is not the cause
F4	cotton, damping x 0.033	+0.19	
F5	wool, bending x 0.1 / x 0.033	+0.52 / +0.33	bending is not the main cause
F6	cotton, bending x 0.033	+0.24	
F7	cotton, tension 5 / 5 / 5 (A)	+0.04	in-plane stiffness is the cause
F8	cotton, tension 15, compression 5, shear 5 (B)	+0.21	
F9	cotton, 5 / 5 / 5, mass x 2 (C)	+0.04	

ID	Fabric test (walk)	Hem lift	Verdict
F10	wool, 5 / 5 / 5	+0.05	chosen: all presets use in-plane stiffness 5



Final hem lift (m) per action, silk / cotton / wool: walk +0.04 / +0.04 / +0.05; run +0.05 / +0.05 / +0.05; slash +0.05 / +0.05 / +0.05; die fwd +0.04 / +0.04 / +0.04; die back +0.06 / +0.05 / +0.05; ride walk +0.03 / +0.03 / +0.03; ride run +0.00 / +0.00 / -0.01.

3. Fit variants and thickness against the original robe 469

Variant	Torso stand-off p50 / p90 cm	Sleeves	Skirt	Result
loose (conform GAP 1.5 cm, LOOSE_K 0.25)	torso 2.2 / 3.3	sleeves 2.4 / 4.0	skirt 8.1 / 17.9	reference
tight (GAP 1.0, LOOSE_K 0, FOLD_MAX 0.4 cm, POSE_GAP 0.6)	torso 1.5 / 2.7	sleeves 1.5 / 2.5	skirt unchanged	0.7-0.9 cm closer: about 0.3 px, hardly visible; shoulders narrower by 1.5-2.5 cm per side
slim cuffs (tight + SLEEVE_CUFF 3.2 cm, CUFF_SPAN 0.25)	-	arm zone ratio 0.66-0.75	-	sleeves slightly thinner than the original robe 469 (0.70-0.84); hand zone only -9% in the walk

Torso band (z 1.02-1.55 m), extra half-width per row in cm, walk:

Height	original robe 469	ours silk (loose)	ours tight
shoulders 1.49-1.56 m	4.5-6.5	4.7-6.9	3.4-4.3
chest 1.1-1.45 m	2.4-3.0	3.2-3.5	3.2-3.8
waist 1.02-1.05 m	1.0-1.6	3.6-3.7	3.4

Arm and hand zone ratio (item pixels / body part pixels):

Action	sleeve 469	sleeve ours	sleeve slim	hand 469	hand ours	hand slim
walk	0.70	0.79	0.66	1.25	2.04	1.86
slash	0.78	0.84	0.72	1.13	1.36	1.30
spell directed	0.72	0.84	0.69	1.25	1.55	1.43
spell area	0.84	0.90	0.75	0.89	1.06	0.99

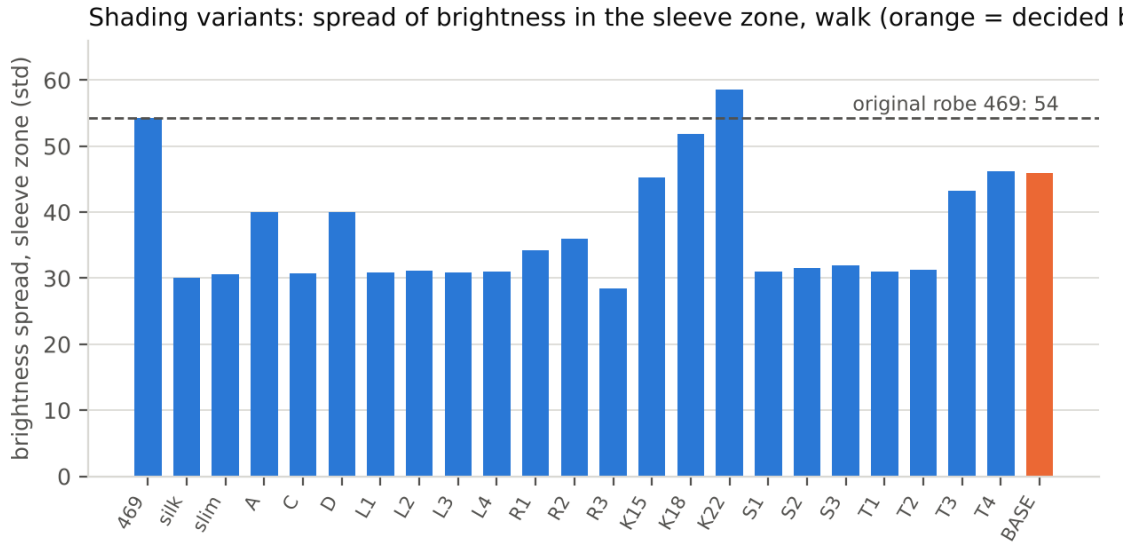
The hand zone in the walk includes the skirt next to the hand (the hands hang against it), which is why it stays high with slim cuffs.

4. Shading variants (sleeve and hand zone, walk)

Brightness spread = standard deviation of luminance. The original robe is painted with dark creases and bright lights; our smooth shell has half the spread. The UO light sits almost on the camera axis (x 0.0012, y -0.757, z 0.653), so thin arms get nearly flat Lambert shading.

ID	Setting	Sleeve std	Hand std	Note
-	original robe 469	54.3	51.9	reference; sleeve p5 = 8, dark share around the hand 56%
-	ours silk, loose (no shading)	30.0	27.5	dark share around the hand 29.5%
-	slim cuffs, no shading	30.6	27.0	
A	inner line 0.6 + outline 0.5	40.0	35.5	a 1 px line round the arm: rejected as unaesthetic
B1	inner line 0.4 + outline 0.3	-	-	rendered, never measured
B2	inner line 0.25 + outline 0.2	-	-	rendered, never measured
C	baked occlusion shading (OCCLUSION 1.0, 8 cm)	30.8	25.7	no effect on arms: baked in the rest pose
D	A + C	40.0	34.0	same as A
L1	own shadow 0.2 (was 0.4)	30.9	27.4	no effect: the light sits on the camera axis
L2	own shadow 0.1	31.2	27.8	no effect
L3	own shadow 0.2 + normal map strength x 2	30.9	27.4	no effect (the material has no normal map)
L4	own shadow 0.2 + texture contrast 1.0	31.0	27.5	no effect
R1	light from the left	34.2	28.5	robe darker overall (mean 96 vs 109)
R2	light from the right	36.0	35.1	robe darker overall
R3	light from above	28.4	25.8	robe darker overall
K15	contrast curve k 1.5 (offline)	45.3	39.5	
K18	contrast curve k 1.8 (offline and in the renderer)	51.8	44.9	renderer: 51.9 / 44.9; closest to 469
K22	contrast curve k 2.2 (offline)	58.6	49.9	too strong
S1	contact shadow from the body 0.6	31.0	27.4	very subtle
S2	contact shadow sleeve-torso 0.6	31.6	27.9	visible depth in the armpit, slightly strong
S3	S1 + S2, width 4 px	32.0	28.3	too dark overall
T1	sleeve shadow 0.4, width 4	31.0	27.4	subtle
T2	sleeve shadow 0.5, width 4	31.3	27.8	subtle
T3	T1 + contrast 1.4	43.2	37.8	
T4	sleeve shadow 0.5 + contrast 1.5	46.2	40.5	dark share around the hand 57%, p5 = 9

ID	Setting	Sleeve std	Hand std	Note
BASE	sleeve shadow 0.4 + contrast 1.5 (decided)	45.9	40.0	base look of robe / skirt; mean sleeve brightness 97 (469: 113), lights 155-163 (469: 180)



5. Decisions, deliverables and code

Deliverables (outside the repository: the robe model and its renders are the user's)

File	Content
robe_silk_full35.vd, robe_cotton_full35.vd, robe_wool_full35.vd	35 actions x 5 directions, the t1 physics, riding with the horse as a collider (without the base look: rendered before it was decided)
robe_silk_tight_full35.vd	the tight fit, same physics
GIFs (8 fps, each action twice, 5 directions)	silk, cotton, wool

New keys and options (all default off or unchanged)

Where	Keys
uo_cloth_sim.py	fabric (silk, cotton, wool, denim, leather), mscale, dscale, bscale, density, goal, pushw, gravity, thick, dist, quality, cq, cull, usenormal, mounted, horse, env / env_margin / env_tent, trace, nocollide
render_uo_layer.py (scene)	uo_contrast, uo_contrast_pivot, uo_shade_sleeve, uo_shade_body, uo_shade_width (README section 4)
uo_make_item.py	scene_of(): base look of kind robe / skirt
item_skin_check.py	feet in their own colour and count (leg px = thigh and shin)
item_gif.py	--crop auto, exact average fps

Lessons from the work

- Background jobs die when the container is reclaimed after idle time: restart the missing jobs and check for DONE markers.
- Do not edit a shell script while it runs (the running copy skipped its last line).
- pkill -f with a pattern that occurs in its own command line kills the shell: kill by PID.
- Blender's per-vertex mass, stiffness and damping are not resolution independent: a 24k vertex mesh needs mass scaled down and the in-plane stiffness lowered.
- Measure what the client shows (item_body_holes.py), not only the 3D preview: the legs outside the robe silhouette are not 'holes' for that metric.

Open issues

- Denim and leather were never simulated.
- The robe is about 1.7 x wider at the hem and 2 cm per side wider at the waist than the original robe 469 (the model has a belt and a wide flare; the pipeline has HEM_FLARE, not used).
- The 35-action .vd files were rendered before the base look (contrast and sleeve shadow) and the slim cuffs: they need a re-render.
- Riding: the robe lies along the legs like trousers (CLOTH_MOUNTED); the horse collider and the horse-cut variant look almost alike in the 3D preview.
- Leg exposure in the falls is the largest remaining one (die forward 385 px through the cloth over 30 frames).
- The sleeve brightness is 97 against 113 and the lights 155-163 against 180: try uo_contrast_pivot 0.55 and a lighter cloth.